

Tutorial 12: Real Insurance Data in R (Student Version)

Theme: Claim Frequency, Severity, Pure Premium and AI-assisted Actuarial Judgement

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Programme: Nurturing Future Risk Managers and Actuaries: An AI-powered Journey for Gifted Secondary Students

Learning Flow

Welcome back, young actuaries! During OSALP, you have learned probability, random variables, distributions, descriptive statistics, simulation, coding and how AI can support learning. Today we connect all of these ideas to one professional question: **how can an insurer use data to understand risk?**

You do not need to become a professional actuary today. Your mission is to ask good questions: What does one row mean? What are we counting? What is the denominator? Is the graph telling a fair story? What could be misleading? If you can explain these in your own words, you are already thinking like a risk analyst.

A. Warm-up: From Insurance Story to Actuarial Data

Learning Flow

Connection to earlier OSALP lessons: We have met uncertainty, expected values, variability, Bernoulli trials, Binomial counts, Poisson counts and descriptive summaries. In this tutorial, these ideas are no longer just textbook formulas. They become tools for reading real insurance data.

Key Idea

In car insurance, the insurer needs to answer three simple but powerful questions:

How often do claims happen? → **Claim Frequency**

How expensive are they when they happen? → **Claim Severity**

How much claim cost should we expect per policy-year? → **Pure Premium**

The main identity is:

$$\text{Pure Premium} = \text{Claim Frequency} \times \text{Claim Severity}.$$

Coding Mission: Opening Scenario

★ Core

Imagine this: An insurance company covers 10,000 cars for one year.

1. Most cars have no claims.
2. Some cars have one claim.
3. A very small number of cars have two or more claims.
4. Most repairs are not huge, but a few claims may be extremely expensive.

Think about it: If you were helping the insurer, what information would you need before suggesting a basic insurance charge?

Worksheet Task: Before any coding

Write one sentence for each question.

1. In plain English, what does *how often* mean in insurance?
2. In plain English, what does *how expensive* mean in insurance?
3. Why might one very large claim matter even if it is rare?

Watch Out! Common Mistake

Pure premium is not the final price paid by customers. It is only the expected claim cost per exposure unit. A real office premium may also include expenses, commissions, profit or risk margin, taxes, levies, reinsurance, regulation and professional judgement.

B. What Data Are We Using?**Learning Flow**

Why real data? Real data is messy. That is exactly why it is useful. It forces us to check rows, columns, units, missing values and whether our conclusions are too quick.

Key Idea

We will use the French Motor Third-Party Liability insurance datasets from the R package `CASdatasets`. We mainly use two tables:

1. `freMTPLfreq`: policy-level data. It contains many policies, exposure time, policy characteristics and the number of claims.
2. `freMTPLsev`: claim-level data. It contains positive claim amounts.
3. `PolicyID`: the common ID tag that allows us to connect the two tables.

Actuary Pro-Tip

The data should be described as historical, de-identified teaching and research data. It is good for learning actuarial thinking, but it is not a current commercial pricing file and should not be used to price Hong Kong motor insurance today.

Coding Mission: Lesson at a Glance

★ Core

Item	Details
Mission	Use R and AI support to analyse claim frequency, severity and pure premium
Your role	Risk analyst / data detective
Coding tool	R, mainly using base R commands
Data	<code>freMTPLfreq</code> and <code>freMTPLsev</code>
Prior knowledge used	probability, Bernoulli/Binomial, Poisson, descriptive statistics, skewness, simulation, graph interpretation
Professional habit	Code, check, interpret, question and document limitations

C. Setup: Let's Get Coding**Learning Flow**

Before analysing, we prepare. In earlier coding activities, you learned that code is not magic. We first install or load what we need, then check whether the data arrived correctly.

Coding Mission: Install and Load the Data Package

★ Core

Task: Run this code once to install the dataset package, then load it.

```
options(timeout = 10000)
install.packages("CASdatasets",
                 repos = "https://cas.uqam.ca/pub/",
                 type = "source")
library(CASdatasets)
```

Worksheet Task: After installing/loading

1. Did R show an error message? If yes, copy the exact first line of the error.
2. If there is no error, write: *The package was loaded successfully.*
3. Ask Gemini only if needed: paste the exact error and ask it to explain the problem in beginner-friendly language.

Coding Mission: Load the Two Tables

★ Core

Task: Load the two datasets and give them short names.

```
data("freMTPLfreq")
data("freMTPLsev")
freq <- freMTPLfreq
sev <- freMTPLsev
```

Worksheet Task: Check your workspace

Run `ls()` in R. Do you see `freq` and `sev`? Write down the two object names you created.

Coding Mission: First Look at the Data

★ Core

Task: Never trust data blindly. Check how many rows and columns there are, the column names and the first few rows.

```
dim(freq)
dim(sev)
names(freq)
```

```
names(sev)
head(freq)
head(sev)
summary(freq)
summary(sev)
```

Worksheet Task: Read the data like a detective

Use your R output to complete the following.

1. `freq` has _____ rows and _____ columns.
2. `sev` has _____ rows and _____ columns.
3. Name one column that appears in both tables: _____.
4. Before calculating anything, what do you think one row of `freq` represents?
5. Before calculating anything, what do you think one row of `sev` represents?

Suggested AI Prompt: Using AI to Read R Output

★ Core

"I ran `dim()`, `names()`, `head()` and `summary()` on two insurance data frames in R. Please help me read the output like a beginner. Tell me what each command checks, what questions I should ask about the rows and columns, and what I should not conclude too quickly."

D. Data Dictionary: What Does One Row Mean?

Learning Flow

The golden rule of data: Before making graphs, know what one row represents. If you confuse a policy row with a claim row, the actuarial calculation becomes wrong.

Coding Mission: Frequency Data Dictionary

★ Core

Variable	Type	Student-friendly meaning
PolicyID	ID tag	A unique label for a policy. It helps connect policy data to claim data.
ClaimNb	Count	Number of claims recorded for that policy.
Exposure	Time at risk	How long the policy was observed, measured in policy-years.
Power	Category	Vehicle power category.
CarAge	Number	Age of the vehicle.
DriverAge	Number	Age of the driver.
Brand	Category	Vehicle brand category.
Gas	Category	Fuel type.
Region	Category	Geographic region.
Density	Number	Population density or related area-density variable.

Coding Mission: Severity Data Dictionary

★ Core

Variable	Type	Student-friendly meaning
PolicyID	ID tag	Shows which policy the claim belongs to.
ClaimAmount	Money	Positive amount paid or recorded for a claim.

Worksheet Task: Policy row versus claim row

Complete the table.

Dataset	One row means	Main outcome	Includes zero-claim policies?
freMTPLfreq			
freMTPLsev			

Watch Out! Common Mistake

A zero-claim policy is still important. It contributes exposure and helps estimate frequency and pure premium. It does not appear as a positive claim amount in the severity table because there is no positive claim amount to record.

E. Mission 1: Exposure and Claim Frequency

Learning Flow

Connection to OSALP statistics: In earlier lessons, you learned that a denominator matters. Here the denominator is not simply the number of policies. It is total exposure, meaning total time at risk.

Key Idea

If policy i has claim count N_i and exposure e_i , the portfolio claim frequency is:

$$\hat{\lambda} = \frac{\sum_i N_i}{\sum_i e_i}.$$

The unit is:

claims per policy-year.

Coding Mission: Look at Exposure Times

★ Core

Task: Summarise and draw a histogram of exposure.

```
summary(freq$Exposure)
hist(
  freq$Exposure,
  breaks = seq(0, max(freq$Exposure, na.rm = TRUE) + 0.05, by = 0.05),
  main = "How Long Are People Insured?",
  xlab = "Exposure (Years)",
  ylab = "Number of Policies",
  col = "lightblue",
  border = "white"
)
```

Worksheet Task: Immediately after the exposure code

Look at the `summary(freq$Exposure)` output.

1. What does exposure 1.00 mean?
2. What does exposure 0.50 mean?
3. Why is it unfair to treat a 3-month policy and a 1-year policy as the same amount of risk time?

Coding Mission: Calculate the Portfolio Frequency

★ Core

Task: Add up all risk time, add up all claims, then divide.

```
total_exposure <- sum(freq$Exposure)
total_claims <- sum(freq$ClaimNb)
portfolio_frequency <- total_claims / total_exposure
total_exposure
total_claims
portfolio_frequency
```

Worksheet Task: Immediately after calculating frequency

Fill in the table using your R output.

Quantity	Formula	Result
Total exposure	$\sum_i e_i$	
Total claims	$\sum_i N_i$	
Portfolio frequency	$\sum_i N_i / \sum_i e_i$	

Now write one sentence: The portfolio frequency means

_____.

Think About It: Unit Checking

Why is saying “claims per policy-year” clearer than simply saying “the claim rate is 5%”?

Suggested AI Prompt: Checking My Frequency Interpretation

★ Core

“My R output gives total exposure, total claims and portfolio claim frequency. Help me write a clear one-sentence interpretation for secondary school students. Please include the unit claims per policy-year and warn me if I accidentally describe it only as a percentage.”

F. Mission 2: Bernoulli and Binomial Thinking

Learning Flow

Connection to earlier probability lessons: You learned that a Bernoulli variable is a Yes/No random variable. Insurance data lets us create one: did this policy have at least one claim?

Coding Mission: Create a “Did They Claim?” Column

★ Core

Task: Create `HasClaim`. It is 1 if the policy had at least one claim, and 0 otherwise.

```
freq$HasClaim <- ifelse(freq$ClaimNb > 0, 1, 0)
mean(freq$HasClaim)
table(freq$HasClaim)
```

Worksheet Task: Immediately after creating `HasClaim`

1. If `ClaimNb` = 0, what is `HasClaim`? _____
2. If `ClaimNb` = 1, what is `HasClaim`? _____
3. If `ClaimNb` = 2, what is `HasClaim`? _____
4. Why does `HasClaim` lose some information compared with `ClaimNb`?

Key Idea

When a result can only be 1 or 0, it behaves like a biased coin flip. In math language:

$$Y_i \sim \text{Bernoulli}(p).$$

If we count how many policies out of n have at least one claim, the simple classroom model is:

$$X \sim \text{Binomial}(n, p).$$

Watch Out! Common Mistake

The Bernoulli/Binomial idea is a simplified classroom model. Real policies do not

all have the same risk. Drivers have different exposure, cars, locations and histories.

Suggested AI Prompt: Claim Indicator versus Claim Count

★ Core

"I created `HasClaim` using `ifelse(ClaimNb > 0, 1, 0)`. Explain the difference between `HasClaim` and `ClaimNb` using examples with 0, 1 and 2 claims. Also explain why `mean(HasClaim)` is not the same as claim frequency per exposure."

G. Mission 3: The Poisson Claim-count Model

Learning Flow

Connection to OSALP distribution learning: Poisson is a counting model. We use it when counting events over time or space, such as emails arriving, calls coming into a centre, or insurance claims occurring during a year.

Key Idea

For a full-year policy, a simple first model is:

$$N_i \sim \text{Poisson}(\lambda).$$

For exposure e_i , the exposure-adjusted idea is:

$$N_i \sim \text{Poisson}(\lambda e_i).$$

A special Poisson rule is:

$$E(N_i) = \text{Var}(N_i) = \lambda.$$

So if the observed variance is much larger than the mean, the simple model may be too simple.

Coding Mission: Compare Real Counts with Poisson Predictions

★★

Intermediate

Task: Use nearly full-year policies first, then compare observed probabilities with Poisson probabilities.

```
freq_1yr <- freq[freq$Exposure >= 0.95 & freq$Exposure <= 1, ]
lambda_1yr <- mean(freq_1yr$ClaimNb)
claim_count_table <- table(freq_1yr$ClaimNb)
poisson_compare <- data.frame(
```

```

ClaimNb = as.numeric(names(claim_count_table)),
Count = as.numeric(claim_count_table)
)
poisson_compare$Observed <- poisson_compare$Count /
  sum(poisson_compare$Count)
poisson_compare$ExpectedPoisson <- dpois(
  poisson_compare$ClaimNb,
  lambda = lambda_1yr
)
poisson_compare

```

Worksheet Task: Immediately after the Poisson comparison table

Use the table to answer.

1. What is `lambda_1yr`? Write it as “average claims per full-year policy”.
2. For `ClaimNb = 0`, is the observed probability close to the Poisson prediction?
3. For `ClaimNb = 1`, is the observed probability close to the Poisson prediction?
4. Write one careful sentence beginning with: “The Poisson model is a useful first benchmark, but ...”

Coding Mission: Draw Real Data versus Poisson Model ★★ Intermediate

Task: Put observed and Poisson probabilities side by side.

```

poisson_small <- poisson_compare[poisson_compare$ClaimNb <= 4, ]
comparison_matrix <- rbind(
  Observed = poisson_small$Observed,
  ExpectedPoisson = poisson_small$ExpectedPoisson
)
barplot(
  comparison_matrix,
  beside = TRUE,
  names.arg = poisson_small$ClaimNb,
  col = c("steelblue", "orange"),
  main = "Real Data vs Poisson Math Model",
  xlab = "Number of Claims",
  ylab = "Probability"
)

```

```

legend(
  "topright",
  legend = c("Real Data", "Poisson Model"),
  fill = c("steelblue", "orange"),
  bty = "n"
)

```

Worksheet Task: Immediately after the Poisson graph

1. Which claim count has the tallest observed bar? _____
2. Does the Poisson model overestimate or underestimate any bar clearly?
Explain briefly.
3. Why is a graph sometimes easier to understand than a table?

Coding Mission: Check the Poisson Mean-Variance Rule ** Intermediate

Task: Compare mean and variance for the full-year group.

```

mean(freq_1yr$ClaimNb)
var(freq_1yr$ClaimNb)

```

Worksheet Task: Immediately after checking mean and variance

Complete the sentences.

1. The observed mean is _____.
2. The observed variance is _____.
3. For a perfect Poisson model, mean and variance should be _____.
4. If variance is much larger, one possible reason is
_____.

Watch Out! Common Mistake

If the model does not match the real world perfectly, that is not a failure. It is a

learning moment. Real drivers may not share one common risk level. This is why actuaries check models instead of trusting formulas blindly.

H. Mission 4: Does Age Matter for Frequency?

Learning Flow

Connection to data interpretation: Earlier lessons warned us not to confuse association with causation. Here we compare groups, but we must use careful wording: “in this historical dataset”, not “age causes claims”.

Coding Mission: Create Age Bands

★ Core

Task: Create driver age bands and vehicle age bands.

```
freq$DriverAgeBand <- cut(
  freq$DriverAge,
  breaks = c(17, 25, 35, 50, 65, Inf),
  right = TRUE,
  labels = c("18-25", "26-35", "36-50", "51-65", "66+")
)
freq$VehicleAgeBand <- cut(
  freq$CarAge,
  breaks = c(-1, 2, 5, 10, Inf),
  right = TRUE,
  labels = c("0-2", "3-5", "6-10", "11+")
)
```

Worksheet Task: Immediately after creating age bands

1. Which age band would a 24-year-old driver enter? _____
2. Which age band would a 40-year-old driver enter? _____
3. Why might changing the age-band cut-points change the results?

Coding Mission: Calculate Frequency by Driver Age

★★ Intermediate

Task: Calculate exposure, claims and frequency for each driver-age group.

```

age_groups <- levels(freq$DriverAgeBand)
frequency_by_age <- data.frame(
  DriverAgeBand = age_groups,
  Policies = as.numeric(table(freq$DriverAgeBand)),
  Exposure = as.numeric(tapply(freq$Exposure, freq$DriverAgeBand, sum)),
  Claims = as.numeric(tapply(freq$ClaimNb, freq$DriverAgeBand, sum))
)
frequency_by_age$Frequency <- frequency_by_age$Claims /
  frequency_by_age$Exposure
frequency_by_age

```

Worksheet Task: Immediately after age-frequency table

Complete one row for each driver-age group using your R output.

Age band	Policies	Exposure	Claims	Frequency
18–25				
26–35				
36–50				
51–65				
66+				

Now write: “In this historical dataset, the group with the highest observed frequency is ...”

Coding Mission: Graph Frequency by Driver Age

★★ Intermediate

Task: Draw a bar chart of claim frequency by age group.

```

barplot(
  frequency_by_age$Frequency,
  names.arg = frequency_by_age$DriverAgeBand,
  col = "orange",
  main = "Claim Frequency by Driver Age",
  xlab = "Driver Age Band",
  ylab = "Claims per Policy-Year"
)

```

Worksheet Task: Immediately after the frequency-by-age graph

1. Which group has the highest bar? _____

2. Which group has the lowest bar? _____
3. Write one possible explanation and one caution. Do not claim causation too quickly.

Actuary Pro-Tip

Observed differences are not automatically pricing rules. They may reflect exposure mix, vehicle mix, geography, claim-reporting patterns, sample size, hidden variables, regulation, fairness concerns or random fluctuation.

I. Mission 5: How Expensive Are the Claims?

Learning Flow

Connection to descriptive statistics: You have learned mean, median, standard deviation and quantiles. Claim severity is a powerful place to use them because insurance costs are often right-skewed.

Coding Mission: Summary of Positive Claim Amounts

★ Core

Task: Calculate mean, median, standard deviation and high percentiles.

```
summary(sev$ClaimAmount)
severity_summary <- data.frame(
  NumberOfClaims = length(sev$ClaimAmount),
  TotalClaimCost = sum(sev$ClaimAmount),
  MeanSeverity = mean(sev$ClaimAmount),
  MedianSeverity = median(sev$ClaimAmount),
  SDSeverity = sd(sev$ClaimAmount),
  Minimum = min(sev$ClaimAmount),
  Maximum = max(sev$ClaimAmount),
  P90 = as.numeric(quantile(sev$ClaimAmount, 0.90)),
  P95 = as.numeric(quantile(sev$ClaimAmount, 0.95)),
  P99 = as.numeric(quantile(sev$ClaimAmount, 0.99))
)
severity_summary
```

Worksheet Task: Immediately after severity summary

Complete the table using R output.

Severity measure	Result
Number of positive claims	
Total claim cost	
Mean severity	
Median severity	
90th percentile	
95th percentile	
99th percentile	
Maximum claim amount	

1. Is the mean larger than the median? What does that suggest?
2. Why might the 99th percentile be useful for an insurer?

Coding Mission: Visualise Claim Costs Below the 99th Percentile

★★

Intermediate

Task: Zoom in so we can see the main body of the distribution.

```
plot_cap <- quantile(sev$ClaimAmount, 0.99)
hist(
  sev$ClaimAmount[sev$ClaimAmount <= plot_cap],
  breaks = 60,
  main = "Claim Costs (Zoomed In)",
  xlab = "Claim Amount",
  ylab = "Number of Claims",
  col = "lightgreen",
  border = "white"
)
mtext(
  "Very large claims are hidden in this figure only",
  side = 3,
  line = 0.5,
  cex = 0.8
)
```


Worksheet Task: Immediately after the zoomed histogram

1. Are most visible claims near the smaller or larger end of the horizontal axis?
2. Did we delete the large claims from the dataset? _____
3. Why did we hide them in this graph only?

Coding Mission: Use a Log Scale**★★ Intermediate****Task:** Plot log10 claim amounts so small and huge claims can be seen together.

```
hist(
  log10(sev$ClaimAmount),
  breaks = 80,
  main = "Claim Costs on a Log Scale",
  xlab = "log10(Claim Amount)",
  ylab = "Number of Claims",
  col = "plum",
  border = "white"
)
```

Worksheet Task: Immediately after the log-scale histogram

1. What does a log scale help us see?
2. Does using log10() mean the original claim amounts were changed permanently? _____
3. Explain why claim amounts are often called right-skewed.

Coding Mission: Optional Normal Q-Q Plot**★★ Intermediate****Task:** Check whether raw positive claim amounts look Normal.

```
qqnorm(
  sev$ClaimAmount,
  main = "Normal Q-Q Plot of Raw Positive Claim Amounts",
  xlab = "Theoretical Normal Quantiles",
```

```

  ylab = "Observed Claim Amounts"
)
qqline(sev$ClaimAmount, col = "red", lwd = 2)

```

Worksheet Task: Immediately after the Q-Q plot

1. Do the points follow the straight line closely? _____
2. Give one reason why a Normal distribution may be unsuitable for raw claim costs.

Watch Out! Common Mistake

Do not say the mean claim is the “typical” claim without explanation. In right-skewed data, a few very large claims can pull the mean upward. The median may describe a middle claim, while the mean is important for expected-cost pricing.

J. Mission 6: Gluing the Data Together

Learning Flow

Connection to coding and data validation: To calculate pure premium, we need policy exposure and claim cost in one table. This is where coding, logic and checking all meet.

Coding Mission: Step 1: Add Up Claim Costs by Policy

★ Core

Task: A policy may have more than one claim. First add up total claim cost per policy.

```

policy_cost_vector <- tapply(sev$ClaimAmount, sev$PolicyID, sum)
policy_cost <- data.frame(
  PolicyID = names(policy_cost_vector),
  TotalClaimCost = as.numeric(policy_cost_vector)
)
head(policy_cost)

```

Worksheet Task: Immediately after aggregating claim cost

1. What does one row of `policy_cost` represent now?
2. Why did we use `tapply()` instead of just using `sev$ClaimAmount` directly?

Actuary Pro-Tip

This tutorial uses base R functions like `tapply()` and `merge()` to reduce package burden. In professional work, actuaries and data scientists may use SQL or `dplyr`, but the idea is the same: aggregate carefully, join carefully, then reconcile totals.

Coding Mission: Step 2: The Left Join

★ Core

Task: Merge policy cost back to the full policy table. Keep all policies, including zero-claim policies.

```
policy_data <- merge(
  freq,
  policy_cost,
  by = "PolicyID",
  all.x = TRUE
)
policy_data$TotalClaimCost[is.na(policy_data$TotalClaimCost)] <- 0
head(policy_data)
```

Worksheet Task: Immediately after the left join

1. Why do some policies have NA before we replace it by 0?
2. Why would deleting zero-claim policies be a serious mistake?
3. In one sentence, explain what `all.x = TRUE` is doing.

Suggested AI Prompt: Visualising a Left Join with AI

★ Core

“I am trying to understand a left join in R using `merge(..., all.x = TRUE)`. Draw a simple text diagram showing what happens when a master table of 5 students is left-joined with a table of 2 library fines. Explain why missing values appear and why we convert them to 0.”

Coding Mission: Step 3: Reconciliation Checks

★★ Intermediate

Task: Check that we did not lose money or claims during the join.

```
sum(policy_data$TotalClaimCost)
sum(sev$ClaimAmount)
sum(policy_data$ClaimNb)
nrow(sev)
```

Worksheet Task: Immediately after reconciliation checks

Complete the table.

Check	Result
Total cost after join: <code>sum(policy_data\$TotalClaimCost)</code>	
Original total claim amount: <code>sum(sev\$ClaimAmount)</code>	
Total claim count after join: <code>sum(policy_data\$ClaimNb)</code>	
Number of rows in severity table: <code>nrow(sev)</code>	
If the first two numbers differ, what should you investigate?	

K. Mission 7: The Grand Finale, Pure Premium**Learning Flow**

The main actuarial chain: Exposure tells us risk time. Frequency tells us how often claims happen. Severity tells us how costly claims are. Pure premium combines both into expected claim cost per policy-year.

Key Idea

At the portfolio level:

$$\text{Frequency} = \frac{\text{Total Claims}}{\text{Total Exposure}}, \quad \text{Severity} = \frac{\text{Total Cost}}{\text{Total Claims}},$$

$$\text{Pure Premium} = \frac{\text{Total Cost}}{\text{Total Exposure}}.$$

Therefore:

$$\text{Frequency} \times \text{Severity} = \text{Pure Premium}.$$

Coding Mission: Calculate Portfolio Metrics

★ Core

Task: Calculate total exposure, claims, cost, frequency, severity and pure premium.

```
portfolio_summary <- data.frame(
  TotalExposure = sum(policy_data$Exposure),
  TotalClaims = sum(policy_data$ClaimNb),
  TotalCost = sum(policy_data$TotalClaimCost)
)
portfolio_summary$Frequency <- portfolio_summary$TotalClaims /
  portfolio_summary$TotalExposure
portfolio_summary$Severity <- portfolio_summary$TotalCost /
  portfolio_summary$TotalClaims
portfolio_summary$PurePremium <- portfolio_summary$TotalCost /
  portfolio_summary$TotalExposure
portfolio_summary
```

Worksheet Task: Immediately after portfolio metrics

Complete the table.

Measure	Unit	Result
Frequency	claims per policy-year	
Severity	currency per claim	
Pure premium	currency per policy-year	

Write one sentence explaining pure premium to a parent or friend who has not studied actuarial science.

Coding Mission: Verify the Formula

★ Core

Task: Check whether frequency times severity equals pure premium.

```
portfolio_summary$Frequency * portfolio_summary$Severity
portfolio_summary$PurePremium
```

Worksheet Task: Immediately after verifying the formula

1. Do the two values agree, apart from rounding? _____
2. Fill in: Frequency $\times$ Severity = _____.
3. Why does the number of claims cancel out algebraically?

Coding Mission: Pure Premium by Driver-age Band

★★ Intermediate

Task: Calculate frequency, severity and pure premium for each age group.

```
age_groups <- levels(policy_data$DriverAgeBand)
pricing_by_age <- data.frame(
  DriverAgeBand = age_groups,
  Policies = as.numeric(table(policy_data$DriverAgeBand)),
  Exposure = as.numeric(tapply(policy_data$Exposure,
    policy_data$DriverAgeBand, sum)),
  Claims = as.numeric(tapply(policy_data$ClaimNb,
    policy_data$DriverAgeBand, sum)),
  TotalCost = as.numeric(tapply(policy_data$TotalClaimCost,
    policy_data$DriverAgeBand, sum))
)
pricing_by_age$Frequency <- pricing_by_age$Claims / pricing_by_age$Exposure
pricing_by_age$Severity <- ifelse(
  pricing_by_age$Claims > 0,
  pricing_by_age$TotalCost / pricing_by_age$Claims,
  NA
)
pricing_by_age$PurePremium <- pricing_by_age$TotalCost /
  pricing_by_age$Exposure
pricing_by_age$FrequencyRelativity <- pricing_by_age$Frequency /
  portfolio_summary$Frequency
pricing_by_age$SeverityRelativity <- pricing_by_age$Severity /
  portfolio_summary$Severity
pricing_by_age$PurePremiumRelativity <- pricing_by_age$PurePremium /
  portfolio_summary$PurePremium
```

```
pricing_by_age
```

Worksheet Task: Immediately after pricing-by-age table

1. Which age group has the highest observed pure premium? _____
2. Is it mainly high because of frequency, severity, or both? Use the table to justify.
3. Which columns help you avoid making a too-simple conclusion?

Coding Mission: Graph Pure Premium by Driver Age

★★ Intermediate

Task: Draw a bar chart of observed pure premium.

```
barplot(
  pricing_by_age$PurePremium,
  names.arg = pricing_by_age$DriverAgeBand,
  col = "pink",
  main = "Expected Cost (Pure Premium) by Driver Age",
  xlab = "Driver Age Band",
  ylab = "Claim Cost per Policy-Year"
)
```

Worksheet Task: Immediately after pure-premium graph

Write a careful three-sentence conclusion.

1. Sentence 1: Describe the main pattern in the graph.
2. Sentence 2: Say whether frequency, severity or both may be driving it.
3. Sentence 3: State one limitation or caution.

K.8 Broader Rating-variable Challenge

Learning Flow

Now think like a real actuarial analyst. So far, we compared pure premium by driver age. But driver age is only one possible way to group policies. In real actuarial work, analysts may compare many rating variables, such as vehicle age, area, vehicle power, claim history, or other risk characteristics.

Your mission is not to find a “perfect” pricing rule. Your mission is to practise the actuarial habit:

compare carefully → check the denominator
 → separate frequency and severity → state limitations.

Actuary Pro-Tip

Why this challenge matters: A group may have a high pure premium for different reasons. It may have more claims per policy-year, larger claims when claims happen, or both. If we only look at the final pure premium, we may miss the real story.

Coding Mission: Prepare an Extra Rating Variable

★★ Intermediate

Task: We already created `DriverAgeBand` and `VehicleAgeBand`. Now we will also create a simple `DensityBand`. This lets students choose from several grouping variables.

Code to run:

```
policy_data$DensityBand <- cut(
  policy_data$Density,
  breaks = c(-Inf, 50, 200, 1000, 5000, Inf),
  right = TRUE,
  labels = c("Very Low", "Low", "Medium", "High", "Very High")
)
table(policy_data$DensityBand)
```

Worksheet Task: Immediately after creating `DensityBand`

1. What does `DensityBand` try to group?

2. Which band has the largest number of policies?
3. Why might a variable related to density be useful in insurance analysis?
4. Why should we still be careful before using it as a final pricing rule?

Coding Mission: Choose One Rating Variable

★★ Intermediate

Task: Choose one rating variable and calculate frequency, severity and pure premium by group.

You may choose one of these:

- "DriverAgeBand"
- "VehicleAgeBand"
- "Region"
- "Power"
- "Gas"
- "DensityBand"

Code to run:

```
rating_variable <- "VehicleAgeBand"
rating_group <- as.factor(policy_data[[rating_variable]])

rating_summary <- data.frame(
  Group = levels(rating_group),
  Policies = as.numeric(table(rating_group)),
  Exposure = as.numeric(tapply(policy_data$Exposure, rating_group, sum)),
  Claims = as.numeric(tapply(policy_data$ClaimNb, rating_group, sum)),
  TotalCost = as.numeric(tapply(policy_data$TotalClaimCost, rating_group,
    sum))
)

rating_summary$Frequency <- rating_summary$Claims / rating_summary$Exposure
rating_summary$Severity <- ifelse(
  rating_summary$Claims > 0,
  rating_summary$TotalCost / rating_summary$Claims,
  NA
)

rating_summary$PurePremium <- rating_summary$TotalCost /
  rating_summary$Exposure
```

```
rating_summary$FrequencyRelativity <- rating_summary$Frequency /
  portfolio_summary$Frequency
rating_summary$SeverityRelativity <- rating_summary$Severity /
  portfolio_summary$Severity
rating_summary$PurePremiumRelativity <- rating_summary$PurePremium /
  portfolio_summary$PurePremium
rating_summary
```

Worksheet Task: Immediately after the broader rating table

Complete the table using your chosen rating variable.

Group	Policies	Exposure	Claims	Cost	Frequency	Pure Premium
1						
2						
3						
4						
5						

1. Which rating variable did you choose?
2. Which group has the highest observed frequency?
3. Which group has the highest observed pure premium?
4. Is the high pure premium mainly driven by frequency, severity, or both? Use the table to justify.
5. Which group has many policies? Which group has only a small number of policies? Why does this matter?

Coding Mission: Graph the Chosen Rating Variable

★★ Intermediate

Task: Draw a bar chart of pure premium for your chosen rating variable.

Code to run:

```
barplot(
  rating_summary$PurePremium,
  names.arg = rating_summary$Group,
```

```

col = "skyblue",
main = paste("Pure Premium by", rating_variable),
xlab = rating_variable,
ylab = "Claim Cost per Policy-Year",
las = 2,
cex.names = 0.8
)

```

Worksheet Task: Immediately after the broader rating graph

Write a careful actuarial interpretation.

1. Describe the main pattern in the graph.
2. Identify whether the highest pure premium group seems to be driven more by claim frequency, claim severity, or both.
3. State one reason why this result may be unstable or misleading.
4. Complete this sentence: "In this historical dataset, the group _____ has a higher observed pure premium, but we should not immediately conclude that _____ causes higher claim cost."

Watch Out! Common Mistake

Do not jump from pattern to pricing rule. If one group has a high observed pure premium, it does not automatically mean the insurer should charge that group more. We should also check sample size, exposure, hidden variables, fairness, regulation and whether the data are still relevant today.

Suggested AI Prompt: Asking AI to Challenge a Rating Conclusion ★★

Intermediate

"I grouped an insurance dataset by a rating variable and calculated exposure, claims, total cost, frequency, severity and pure premium. One group has the highest pure premium. Please help me check whether this is mainly driven by frequency, severity or both. Also challenge my conclusion by asking what limitations, sample-size issues,

hidden variables or fairness concerns I should consider."

Coding Mission: Optional: Try a Second Rating Variable *** Challenge

Task: Change only the first line of the earlier code.

For example:

```
rating_variable <- "Region"
```

or:

```
rating_variable <- "DensityBand"
```

Then run the same calculation again.

Question: Does the actuarial story change when you choose a different rating variable?

Worksheet Task: Compare two rating variables

If you tried two rating variables, complete this short comparison.

	First rating variable	Second rating variable
Variable chosen		
Highest pure pre-		
mium group		
Main driver: fre-		
quency, severity		
or both?		
One limitation		

Which rating variable gave a clearer actuarial story? Explain briefly.

Actuary Pro-Tip

Pure premium is an expected claim cost measure, not a complete pricing decision. A commercial premium may include expenses, commissions, profit or risk margin, taxes, levies, capital support, reinsurance, regulatory constraints, fairness, strategy and professional judgement.

L. AI-supported Investigation Protocol

Learning Flow

How to use AI like a pro: Google Gemini or another AI assistant can help explain code, interpret output, debug errors and challenge reasoning. It should not replace your thinking.

Key Idea

A strong AI-assisted workflow is:

1. Try the R code yourself.
2. Run the code and read the output or error.
3. Ask AI to explain the code, error or output.
4. Check whether the AI explanation matches the actuarial definitions.
5. Fix or improve the code.
6. Write your own interpretation.

Watch Out! Common Mistake

A weak prompt asks: “What is the answer?” A stronger prompt asks: “Here is my code, here is my output, and here is my interpretation. Please check my reasoning, explain any mistake, and ask me one question to improve my analysis.”

Coding Mission: Gemini Prompt Menu for Students

★ Core

Copy, adapt and improve these prompts during the worksheet.

1. **Before coding:** *“I am about to analyse an insurance dataset in R. Help me list the questions I should ask before calculating anything: what does one row mean, what variables are important, and what mistakes should I avoid?”*
2. **After an error message:** *“Here is my R code and the exact error message. Explain the likely cause in simple language, then show a corrected version using base R where possible. Also tell me how I can check that the corrected code makes sense.”*
3. **After a table output:** *“Here is my R output table. Help me interpret each column in actuarial language. Do not just repeat the numbers. Explain what*

the numbers mean for claim frequency, claim severity or pure premium.”

4. **For understanding joins:** *“I am using a left join to merge two tables. Draw a simple text diagram explaining why missing values happen when some rows in the first table do not exist in the second table.”*
5. **After a graph:** *“I made this graph from insurance data. Help me describe the main pattern, one possible explanation and one caution. Please avoid claiming causation too quickly.”*
6. **For checking units:** *“Please check whether I am using the correct unit: claims per policy-year, currency per claim, or currency per policy-year. Explain why the unit matters.”*
7. **For professional judgement:** *“Challenge my actuarial conclusion. What could make my interpretation misleading? Consider exposure, sample size, missing variables, historical data and fairness.”*

M. Consolidated Student Worksheet

Coding Mission: Part A: Understanding the Data

★ Core

1. What is the unit of observation in `freMTPLfreq`?
2. What is the unit of observation in `freMTPLsev`?
3. Which variable links the two datasets?
4. What does exposure 0.50 mean?
5. Why does the frequency dataset contain more observations than the severity dataset?

Coding Mission: Part B: Frequency and Distributions

★ Core

1. In your own words, what does portfolio frequency mean?

2. What is the difference between `HasClaim` and `ClaimNb`?
3. How does `HasClaim` connect to Bernoulli random variables?
4. How does claim count connect to the Poisson distribution?
5. Under a Poisson model, what relationship should mean and variance have?
6. If variance is much greater than mean, what might that suggest?

Coding Mission: Part C: Severity and Skewness

★★ Intermediate

1. Is the mean claim cost usually higher or lower than the median in this dataset? Why?
2. Why do insurance companies care about high percentiles such as the 95th or 99th percentile?
3. What does right-skewed mean in the context of claim amounts?
4. Give one reason why raw positive claim amounts may not be well modelled by a Normal distribution.

Coding Mission: Part D: Pure Premium and Actuarial Judgement

★★

Intermediate

Complete the formulas:

Frequency = $\frac{\text{ClaimNb}}{\text{Exposure}}$,

Severity = $\frac{\text{ClaimAmount}}{\text{ClaimNb}}$,

Pure Premium = $\text{Frequency} \times \text{Severity}$.

1. Why should frequency times severity equal pure premium?
2. If one age group has high pure premium, how can you check whether it is driven by frequency, severity or both?
3. Why should observed group differences not immediately become final pricing rules?

N. Optional Extension Missions

Coding Mission: Extension 1: Simulate Bernoulli Claims ★★ Intermediate

Task: Use random numbers to simulate whether 10,000 virtual policies have a claim.

```
set.seed(2026)
n <- 10000
p <- mean(freq$HasClaim)
u <- runif(n)
simulated_has_claim <- ifelse(u < p, 1, 0)
mean(simulated_has_claim)
```

Worksheet Task: After Bernoulli simulation

1. Is the simulated mean close to p? _____
2. Why is it not exactly equal every time?

Coding Mission: Extension 2: Simulate Poisson Claim Counts ★★

Intermediate

Task: Simulate annual claim counts using the portfolio frequency.

```
set.seed(2026)
lambda <- portfolio_summary$Frequency
simulated_claims <- rpois(
```



```

n = 10000,
lambda = lambda
)
mean(simulated_claims)
var(simulated_claims)
table(simulated_claims)

```

Worksheet Task: After Poisson simulation

1. Are the simulated mean and variance close? _____
2. How does this connect to the Poisson mean-variance rule?

Coding Mission: Extension 3: Empirical Severity Simulation*** Advanced

Task: Resample observed claim amounts without introducing a new severity distribution.

```

set.seed(2026)
simulated_severity <- sample(
  sev$ClaimAmount,
  size = 10000,
  replace = TRUE
)
mean(simulated_severity)
median(simulated_severity)
quantile(simulated_severity, c(0.50, 0.90, 0.99))

```

Worksheet Task: After empirical severity simulation

1. Does the simulated severity still look right-skewed? _____
2. Why might resampling old claims be risky for future pricing?

Coding Mission: Extension 4: Simple Aggregate-loss Simulation

Advanced

Task: Combine Poisson frequency with empirical severity resampling.

```
set.seed(2026)
n_policies <- 10000
lambda <- portfolio_summary$Frequency
claim_counts <- rpois(n_policies, lambda)
total_number_claims <- sum(claim_counts)
simulated_claim_amounts <- sample(
  sev$ClaimAmount,
  size = total_number_claims,
  replace = TRUE
)
aggregate_loss <- sum(simulated_claim_amounts)
aggregate_loss
aggregate_loss / n_policies
```

Worksheet Task: After aggregate-loss simulation

Explain in one short paragraph how frequency and severity combine to produce aggregate loss.

Coding Mission: Extension 5: Professional-looking Graphs with ggplot2

*** Advanced

Task: If you finish early, ask Gemini to help you rewrite one base R graph using ggplot2. Do not just copy. Ask it to explain `aes()`, `geom_bar()` and the idea of a chart theme. **Suggested AI prompt:** “I have created a basic bar chart of

insurance pure premium by driver-age band using base R. I want to recreate this as a professional chart using ggplot2. I have never used ggplot2. Guide me step by step and explain each part of the code.”

O. Final Takeaway

Key Idea

This tutorial turns insurance data into actuarial thinking.

- **Exposure** tells us how much risk time is observed.
- **Frequency** tells us how often claims occur.

- **Severity** tells us how costly claims are when they occur.
- **Pure premium** combines frequency and severity into expected claim cost per policy-year.
- **AI support** can explain code, debug errors and challenge reasoning, but the analyst is still responsible for definitions, units, assumptions and limitations.

End of Tutorial 12